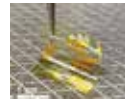




Twittering with Paralysis

Patient unable to move due to a nervous system disorder shared a tweet by thinking about it. <https://digit.in/jan22-41>



Lightweight solar panels

Stanford scientists have developed a new photovoltaic material for lightweight solar panels. <https://digit.in/jan22-42>

WHAT'S YOUR TYPE?

Why do humans have different blood groups

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owadays, a number of lives are saved due to blood transfusion, but before the discovery of blood

types, countless lives were lost. Blood transfusion was done by transfusing animal blood into humans which resulted in vomiting and instant death. British physician James Blundell, stressed by the death of female patients due to blood loss during childbirth, was convinced that blood transfusion would save their lives. He was the first one who insisted that humans should receive blood from another human, not a different species. Blundell injected blood in the arm of a person bleeding to death who felt better after the procedure but died two days later. The experience, nevertheless, encouraged Blundell to continue his experiment. In total he performed 10 blood transfusions of which four patients survived.

As to why the survival rate of transfusion was low was later found out to be clumps of blood. The scientist noticed that some red blood cells stuck to each other but they did not pay much attention to that detail dismissing it as some sort of pathology



not worth investing too much time into. It was not until Karl Landsteiner in 1909 studied the reason behind the clumps and discovered the different types of blood. The formation of clumps was due to transfusion between different blood types. It was the groundbreaking study by Landsteiner and his students which brought to light that fatalities occurred only when blood transfusion was done with the blood of different groups. For this discovery, Karl Landsteiner received the Nobel Prize for Physiology or Medicine in 1930.

There are four primary classifications of blood type and their existence is to fight off critical illness and to boost immunity. Evolution has played a pivotal part in humans having different blood groups.

Blood group A is chronologically the oldest found blood type, and it existed before the homo sapiens evolved from their primate ancestors. Blood type B was discovered some 3.5 million years ago, due to genetic mutations that shifted one of the sugar molecules that sit on the surface of RBC (red blood corpuscles). Due to further mutations, the sugar gene disappeared from the RBC creating the blood type O, which had neither A nor B adaptation of the sugar. Later, the rare blood type AB was found which is covered with both A and B sugar molecules.

Up until the study, it was believed that all human blood is similar. Then it was found that there are different types of blood, A, B, AB, O, and their positive and negative antigens.